

“It would be unwise of you, since you are obliged to play, not to risk your life to win three lives at a game in which there is an equal chance of winning and losing. But here there is an infinity of happy life to be won ... and what you are staking is finite. ... And thus, since you are obliged to play, you must be renouncing reason if you hoard your life rather than risk it for an infinite gain, which is just as likely to occur as a loss...”

This rule of belief led Pascal to a famous argument for belief in God.

Our question is: how might Pascal argue that believing in God has higher expected utility than nonbelief?

First, he emphasizes that “there is an equal chance of gain and loss” — an equal chance that God exists, and that God does not exist. This means that we should assign each a probability of $1/2$.

Second, he says that in this case the amount to be won is infinite. We can represent this by saying that the utility of belief in God if God exists is ∞ .

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